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Perfluorinated chemicals in blood of residents in Catalonia (Spain) in relation to age and gender: A pilot study

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Abstract

Fluorinated organic compounds (FOCs) are a group of chemicals widely used as surfactants, lubricants, polymers, and fire-fighting foams. Recent studies have shown the ubiquitous distribution of FOCs in the environment, wildlife, and humans. We here report the results of a pilot study conducted to provide preliminary data on the levels of 13 FOCs in the blood of 48 residents in Catalonia, Spain, in relation to gender and age (25±5 and 55±5 years). The highest mean concentration was obtained for perfluorooctane sulfonate (PFOS, 7.64 ng/ml), followed by perfluorohexane sulfonate (PFHxS, 3.56 ng/ml) and perfluorooctanoic acid (PFOA, 1.80 ng/ml). Four other FOCs showed mean levels between 0.30 and 0.44 ng/ml, whereas those of the remaining 6 compounds were below the detection limit. Regarding gender, the blood levels of PFHxS and PFOA were significantly higher ($p<0.05$) in men than in women, while differences according to age were only noted for PFHxS ($p<0.05$) and perfluorooctane sulfonamide (PFOSA) ($p<0.001$), for which the levels were higher in the younger (25±5 years) group of subjects. A significant correlation between PFOS levels and those of the remaining detected FOCs (except PFDA) was found. In general terms, the current FOC concentrations were lower than those found in recent studies concerning levels of these chemicals in human blood and serum of subjects from different countries.
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Keywords: Perfluorinated compounds; Biomonitoring; Whole blood; General population

1. Introduction

Fluorinated organic compounds (FOCs) form a diverse group of chemicals with surface-active properties, which have been manufactured for over 50 years. Because of their unique properties of repelling both water and oil, FOCs have been widely used as surfactants and surface protectors in carpets, leather, paper, food containers, fabric, and upholstery (US EPA, 2000; Giesy and Kannan, 2002; Beach et al., 2006). These compounds have also been used in fire-fighting-foams, floor polishes, and shampoos (US EPA, 2000). Although FOCs have been manufactured for over 50 years, research investigating the potential toxicity and the environmental fate of these chemicals has been quantitatively less important than that concerning chlorinated and brominated organic compounds. Because much of the FOCs were incorporated into polymers, they were perceived as more biologically inert and therefore, less likely to have an impact on human health or the environment (Giesy and Kannan, 2001).

Among the different classes of FOCs, in recent years perfluorooctanesulfonate (PFOS) and perfluorooctanoic acid (PFOA), especially PFOS, have received worldwide attention, being the subject of a number of investigations. PFOS is the end product of degradation of various sulfonated fluorochemicals. It is stable in the environment and resists hydrolysis, photolysis and biodegradation, and it has been widely found in environmental samples and wildlife tissues (Giesy and Kannan, 2001, 2002; Kannan et al., 2001; 2002; Taniyasu et al., 2003; Martin et al., 2004; So et al., 2004; Holmström et al., 2005). PFOA, as well as other FOCs such as perfluorhexanesulfonate (PFHS), or perfluorooctanesulfonamide (PFOSA) has also been found in

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the environment, although at lower concentrations and frequencies than those of PFOS (Kannan et al., 2005). The environmental persistence, the potential for long-term environmental effects, as well as the finding of PFOS in blood of non-occupationally exposed humans, raised a notable concern about PFOS and related compounds (US EPA, 2000). As a direct consequence, the main producer of PFOS-derived chemicals, 3M Company (USA), announced in May 2000 that they would phase out the production of these substances, which ceased at the end of 2002 (Lau et al., 2003). However, other markets still continue with the production, whereas PFOA remains also in production by other manufacturers.

Studies concerning toxicity of FOCs in mammals have been mainly focused on PFOS and PFOA. The toxic potentials of these compounds were recently reviewed (3M, 2003; Kennedy et al., 2004). Subchronic exposure to PFOS led to effects on cell-to-cell communication and cellular energetic metabolism, disrupting mitochondrial metabolism, and inducing peroxisomal proliferation (Berthiaume and Wallace, 2002). In turn, PFOA has the potential to induce peroxisomal β-oxidation in the liver of male rats (Berthiaume and Wallace, 2002; Lau et al., 2004). Moreover, developmental toxicity of both PFOS and PFOA has been described in rodent models (Lau et al., 2004; Fuentes et al., 2006).

Although accumulation and trends of FOCs are largely unknown, it is evident that unlike the classical more lipophilic persistent organic pollutants such as dioxins and furans (PCDD/PCDFs), or PCBs, FOCs do not typically accumulate in lipids. In humans, exposure levels and pathways leading to the presence of FOCs can be better characterized by monitoring these chemicals in blood. In recent years, information concerning the levels of various FOCs in human blood has been reported for a number of countries (Kannan et al., 2004; Calafat et al., 2006a,b; Kärrman et al., 2006a,b; Harada et al., 2007; Yeung et al., 2006). The present study has been the first monitoring survey of FOCs in a Spanish population. The aim of this investigation was to determine the levels and prevalence of 13 FOCs in whole blood of residents in Catalonia (NE Spain). The results were compared with those previously reported in other countries. The influence of age and gender on the blood levels of FOCs was also assessed.

2. Materials and methods

2.1. Sampling

Forty-eight samples of whole blood corresponding to healthy adult individuals living in Tarragona County (Catalonia, Spain) for at least the last 5 years, were obtained from out-patient departments of the Hospital "Joan XXIII" (Tarragona). All participants were over 20 years old. Twenty-four men and 24 women were included in the study. Each gender group was divided into two subgroups according to age: 25 ± 5 and 55 ± 5 years. Sampling period was May 2006. Blood samples of about 5 ml were collected (in EDTA) from each subject by venipuncture and kept into thoroughly cleaned polypropylene (PP) tubes. The PP tubes and their tops were previously rinsed with methanol. All samples were immediately frozen at −20 °C until analyzed.

2.2. Chemicals

Ammonium acetate (>99%, p.a. for HPLC) was purchased from Fluka (Steinheim, Germany), formic acid (98–100%) from Scharlau (Barcelona, Spain), and methanol (HPLC) from Labscan (Dublin, Ireland). Laboratory produced ultra pure water was used. Ammonium hydroxide (25% in water) and sodium acetate were purchased from E. Merck (Darmstadt, Germany). Perfluorobutane sulfonates (PFBuS) tetrabutylammonium salt (>98%), perfluorooctane sulfonates (PFOS) potassium salt (>98%), perfluorodecanoic acid (PFDA, >97%), and perfluorohexanoic acid (PFHxA >97%) were purchased from Fluka. Perfluoroheptanoic acid (PFHpA, 99%), perfluorononanoic acid (PFNA, 97%) perfluorooctanoic acid (PFOA, 96%), perfluorodecane sulfonate (PFDS) ammonium salt (25 wt.% in 2-butoxyethanol (37%) in water), perfluoroundecanoic acid (PFUnDA, 95%), and perfluorotetradecanoic acid (PFTDA, 97%) were purchased from Aldrich (Steinheim, Germany and Milwaukee, WI, USA). Perfluorooctane sulfonamide (PFOSA, 97%) and 7*H*-PFHpA (98%) were purchased from ABCR (Karlsruhe, Germany). 1*H*,1*H*,2*H*,2*H*-PFOS (THPFOS, purity unknown), and perfluorohexane sulfonates (PFHxS, 98%) were purchased from Interchim (Montlucon, France). $^{13}C_4$-labeled PFOA, $^{13}C_4$-labeled PFOS, and $^{13}C_5$-PFNA were from Wellington Laboratories (Guelph, Ontario, Canada).

2.3. Analyses

Whole blood samples were extracted using weak anion exchange, solid-phase extraction (Waters Oasis® WAX), based on Taniyasu et al. (2005). Extraction standards ($^{13}C_4$-PFOA and $^{13}C_4$-PFOS) and 2 ml formic acid/water (1:1) were added to 0.5 ml whole blood. The solution was sonicated for 15 min and centrifuged at 30 000 ×*g* for 30 min. The supernatant was extracted on a WAX column (200 mg/2 ml) previously conditioned with 2 ml methanol and 2 ml water. FOCs were eluted with 1 ml 2% ammonium hydroxide in methanol, after washing the column with 2 ml 40% methanol. Extracts were filtered using a

Table 1
Perfluorinated chemicals included in the selected ion monitoring method including quantification ion and fragmentation

Compound	Abbreviation	Molecular formula	Quantification ion (m/z)	Fragmentation voltage (V)
Perfluorobutanesulfonate	PFBuS	$C_4F_9SO_3^-$	299	95
Perflurohexanesulfonate	PFHxS	$C_6F_{13}SO_3^-$	399	130
Perfluorooctanesulfonate	PFOS	$C_8F_{17}SO_3^-$	499	150
$^{13}C_4$-perfluorooctanesulfonate	$^{13}C_4$-PFOS	1,2,3,4-^{13}C-$C_4F_{17}SO_3^-$	503	150
1*H*,1*H*,2*H*,2*H*-perfluorooctanesulfonic acid	THPFOS	$C_8F_{13}H_4SO_3H$	427	110
Perfluorodecanesulfonate	PFDS	$C_{10}F_{21}SO_3^-$	599	150
Perfluorohexanoic acid	PFHxA	$C_5F_{11}CO_2H$	269	65
Perfluoroheptanoic acid	PFHpA	$C_6F_{13}CO_2H$	319	70
Perfluorooctanoic acid	PFOA	$C_7F_{15}CO_2H$	369	75
$^{13}C_4$-perfluorooctanoic acid	$^{13}C_4$-PFOA	1,2,3,4-^{13}C-$C_3F_{15}CO_2H$	372	75
Perfluorononanoic acid	PFNA	$C_8F_{17}CO_2H$	419	75
$^{13}C_5$-perfluorononanoic acid	$^{13}C_5$-PFNA	1,2,3,4,5-^{13}C-$C_3F_{17}CO_2H$	423	75
Perfluorodecanoic acid	PFDA	$C_9F_{19}CO_2H$	469	83
Perfluoroundecanoic acid	PFUnDA	$C_{10}F_{21}CO_2H$	519	85
Perfluorotetradecanoic acid	PFTDA	$C_{13}F_{27}CO_2H$	669	93
Perfluorooctanesulfonamide	PFOSA	$C_8F_{17}SO_2NH_2$	498	113

Table 2
Individual concentrations (ng/ml) of 13 perfluorinated chemicals in whole blood of subjects living in Tarragona County (Catalonia, Spain)

ID	Sex	Age	PFBuS	PFHxA	PFHpA	PFHxS	THPFOS	PFOA	PFNA	PFOS	PFOSA	PFDA	PFUnDA	PFDS	PFTDA
1	Male	26	ND	ND	ND	3.68	ND	1.58	ND	11.6	0.36	ND	ND	ND	ND
2	Male	25	ND	ND	ND	3.33	ND	1.09	ND	4.36	0.34	ND	ND	ND	ND
3	Male	24	ND	ND	ND	6.58	ND	1.48	ND	3.68	0.25	ND	ND	ND	ND
4	Male	23	ND	ND	ND	1.05	ND	1.06	ND	0.88	ND	ND	ND	ND	ND
5	Male	23	ND	ND	ND	3.93	ND	1.84	ND	7.40	0.32	ND	ND	ND	ND
6	Male	27	ND	ND	ND	7.29	ND	3.13	0.72	8.15	ND	0.7	ND	ND	ND
7	Male	28	ND	ND	ND	5.94	ND	2.27	0.68	13.9	0.78	0.71	0.78	ND	ND
8	Male	23	ND	ND	ND	3.24	ND	1.76	0.43	9.04	0.26	ND	ND	ND	ND
9	Male	23	ND	ND	ND	7.14	ND	1.02	ND	10.8	0.32	ND	ND	ND	ND
10	Male	24	ND	ND	ND	20.0	ND	2.33	ND	16.2	0.53	ND	ND	ND	ND
11	Male	22	ND	ND	ND	5.27	ND	1.85	ND	8.68	0.29	ND	ND	ND	ND
12	Male	20	ND	ND	ND	5.51	ND	2.35	0.58	6.10	ND	ND	ND	ND	ND
13	Male	54	ND	ND	ND	3.32	ND	1.97	ND	8.47	0.53	ND	0.74	ND	ND
14	Male	56	ND	ND	ND	2.30	ND	1.27	ND	5.73	0.46	ND	ND	ND	ND
15	Male	56	ND	ND	ND	2.03	ND	1.56	ND	6.85	0.88	ND	ND	ND	ND
16	Male	60	ND	ND	ND	2.89	ND	3.11	0.81	9.82	0.37	0.71	0.53	ND	ND
17	Male	57	ND	ND	ND	1.21	ND	3.06	0.51	2.71	0.20	ND	ND	ND	ND
18	Male	57	ND	ND	ND	1.98	ND	2.72	0.68	11.6	0.29	0.65	0.54	ND	ND
19	Male	50	ND	ND	ND	3.47	ND	2.19	0.41	13.4	0.47	ND	0.46	ND	ND
20	Male	56	ND	ND	ND	3.28	ND	0.87	ND	4.10	0.23	ND	ND	ND	ND
21	Male	50	ND	ND	ND	3.25	ND	1.76	1.29	7.73	0.64	0.57	0.51	ND	ND
22	Male	60	ND	ND	ND	2.63	ND	2.80	ND	6.66	0.33	ND	ND	ND	ND
23	Male	51	ND	ND	ND	7.85	ND	2.93	0.72	13.7	0.77	0.61	0.66	ND	ND
24	Male	53	ND	ND	ND	2.77	ND	2.58	0.83	12.0	0.91	ND	0.41	ND	ND
25	Female	26	ND	ND	ND	3.85	ND	1.17	0.53	8.84	0.47	ND	0.84	ND	ND
26	Female	28	ND	ND	ND	1.12	ND	1.36	ND	0.76	0.23	0.44	ND	ND	ND
27	Female	24	ND	ND	ND	5.12	ND	2.43	0.52	7.23	0.41	0.62	ND	ND	ND
28	Female	20	ND	ND	ND	1.70	ND	1.42	0.42	6.62	0.50	ND	ND	ND	ND
29	Female	23	ND	ND	ND	5.16	ND	1.92	0.49	3.03	0.17	ND	ND	ND	ND
30	Female	26	ND	ND	ND	3.08	ND	2.28	0.48	7.21	ND	ND	0.40	ND	ND
31	Female	23	ND	ND	ND	2.51	ND	0.99	ND	3.34	0.25	ND	ND	ND	ND
32	Female	30	ND	ND	ND	3.48	ND	1.40	0.42	5.04	0.38	ND	ND	ND	ND
33	Female	23	ND	ND	ND	2.33	ND	1.31	ND	2.00	ND	ND	ND	ND	ND
34	Female	26	ND	ND	ND	2.41	ND	1.39	ND	6.09	0.39	ND	ND	ND	ND
35	Female	27	ND	ND	ND	4.28	ND	1.57	1.49	9.19	ND	ND	0.53	ND	ND
36	Female	27	ND	ND	ND	3.74	ND	1.24	0.57	9.24	0.18	ND	0.64	ND	ND
37	Female	59	ND	ND	ND	0.65	ND	1.63	0.38	7.74	0.33	ND	0.48	ND	ND
38	Female	51	ND	ND	ND	2.44	ND	1.19	ND	9.27	0.61	ND	ND	ND	ND
39	Female	52	ND	ND	ND	1.82	ND	0.99	0.81	3.99	0.32	ND	ND	ND	ND
40	Female	52	ND	ND	ND	1.58	ND	1.23	0.47	5.26	1.10	ND	ND	ND	ND
41	Female	55	ND	ND	ND	1.54	ND	0.79	ND	5.16	0.54	ND	ND	ND	ND
42	Female	50	ND	ND	ND	1.68	ND	2.00	0.64	8.81	0.65	ND	0.50	ND	ND
43	Female	54	ND	ND	ND	2.96	ND	1.28	ND	7.47	0.24	ND	0.48	ND	ND
44	Female	53	ND	ND	ND	2.84	ND	1.58	0.8	12.0	0.31	ND	0.60	ND	ND
45	Female	54	ND	ND	ND	0.99	ND	1.71	ND	7.98	0.42	ND	0.48	ND	ND
46	Female	59	ND	ND	ND	2.54	ND	3.09	0.45	9.67	1.35	ND	0.43	ND	ND
47	Female	54	ND	ND	ND	1.26	ND	2.03	ND	4.91	0.76	ND	ND	ND	ND
48	Female	55	ND	ND	ND	2.13	ND	1.67	0.49	12.6	0.60	ND	0.60	ND	ND
LOD			0.73	1.00	0.78	0.14	1.10	0.66	0.41	0.09	0.16	0.48	0.40	0.58	0.44
Mean			ND	ND	ND	3.56	ND	1.80	0.44	7.64	0.42	0.30	0.34	ND	ND
SD			–	–	–	2.97	–	0.66	0.29	3.54	0.27	0.15	0.19	–	–
Median			–	–	–	2.92	–	1.65	0.41	7.60	0.35	0.24	0.20	–	–
Min.			–	–	–	0.65	–	0.79	ND	0.76	ND	ND	ND	–	–
Max.			–	–	–	20.0	–	3.13	1.49	16.2	1.35	0.71	0.84	–	–

ND: Not detected.

0.2 μm PP-filter and the final volume was adjusted to 500 μl. A syringe standard, $^{13}C_5$-PFNA, was immediately added before injection.

Analyses were performed using an Agilent 1100 HPLC system (Agilent, Waldbronn, Germany) coupled to an HP 1100 mass spectrometric detector (MSD, Waldbronn, Germany) with an atmospheric electrospray interface operating in negative ion mode. Separation was performed on a Discovery HS C18 (50 mm length, 2.1 mm i.d., 3 μm particles, 120 Å pore size) column with a guard column of the same material (20 mm length, 2.1 mm i.d., 3 μm particles, 120 Å pore size) (Supelco, Bellefonte, PA, USA). Both columns were kept at 40 °C. An extra guard column (HyperCarb graphitic carbon, 4 mm length, 10 mm i.d., 5 μm particle size) (Termo Hypersil-Keystone, Bellefonte, PA, USA) was inserted between the pump and the injector to remove any fluorochemicals originated from the HPLC system. Injection volume was 10 μl and the flow rate was set to 300 μl/min. The mobile phases consisted of 2 mM ammonium acetate in methanol and 2 mM ammonium

Table 3
Mean concentrations (ng/ml) of 13 perfluorinated chemicals in whole blood of subjects living in Tarragona County (Catalonia, Spain) classified according to two age groups

	Age 25±5 years	Age 55±5 years	p
PFBuS	ND	ND	
PFHxA	ND	ND	
PFHpA	ND	ND	
PFHxS	4.65±3.70	2.48±1.39	<0.05
THPFOS	ND	ND	
PFOA	1.68±0.55	1.92±0.74	NS
PFNA	0.41±0.29	0.47±0.30	NS
PFOS	7.06±3.86	8.23±3.15	NS
PFOSA	0.29±0.18	0.55±0.18	<0.001
PFDA	0.30±0.15	0.31±0.15	NS
PFUnDA	0.30±0.19	0.39±0.18	NS
PFDS	ND	ND	
PFTDA	ND	ND	

ND: not detected. NS: not statistically significant ($p>0.05$).

acetate in water. The mobile phase gradient started at 35% methanol followed by a 20 min ramp to 90%, a 2 min hold followed by a 3 min washing sequence with 100% methanol, and then reverting to initial conditions allowing 7 min stabilization time. The MS settings were as follows: nitrogen nebulizer gas temperature 350 °C, nebulizer gas pressure 20 psi, nitrogen drying gas flow 13 ml/min, and capillary voltage 3500 V. Selected ion monitoring (SIM) was used to measure the ions at fragmentation voltage (Table 1).

2.4. Quantification and quality assurance

Quantification was performed using the internal standard method. Standards dissolved in 35% methanol in water were used to calculate relative response factors (RRFs). $^{13}C_4$-PFOS was used as an extraction standard for the sulfonates and PFOSA, whereas $^{13}C_4$-PFOA was used for the carboxylates. The syringe standard, $^{13}C_5$-PFNA, was used to monitor the recovery of the extraction standards. Recoveries of 74–111% and 76–102% were achieved for $^{13}C_4$-PFOA and $^{13}C_4$-PFOS respectively. Recovery experiments showed similar recoveries for the extraction standards (84–87%) and the target compounds (86–110%), except for PFOSA. The method validation using ^{13}C-labelled standards has been recently reported (Kärrman et al., in press). The limit of detection (LOD) was determined as three times the signal to noise ratio of the lowest concentration of a 5–7 point calibration curve. Procedural blank level trace was found only for PFOA, at a concentration below the limit of detection. Ion suppression was examined by calculating the ratio between a standard solution and the same standard solution added to extracted blood. All ratios were between 0.94–1.18 indicating no ion enhancement or suppression for all compounds except PFOSA. External quality assurance was performed by successful participation (z-scores <2) in the 1st and 2nd interlaboratory study on perfluorinated chemicals (van Leeuwen et al., in press).

2.5. Data analysis

Statistical significance of the data was computed by the Student's t test. The Levene proof was previously executed to determine whether the data followed a normal distribution. A probability of 0.05 or less ($p<0.05$) was considered as significant. Pearson's correlation coefficients were used to evaluate the relationships among the analyzed FOCs. Statistical analysis was carried out using the SPSS-12.0 Statistical Software Package.

To establish pattern similarities among the different analyzed FOCs, a self-organizing map (SOM) was applied to the whole blood samples. This is a kind of unsupervised artificial neural network, originally created by Kohonen (1982). SOM is a powerful "data-mining"-based chemometric tool generally used to classify large amounts of data (Nadal et al., 2004a). The basis for data treatment and analysis were recently reported (Ferré-Huguet et al., 2006). The Kohonen neural network consists of two layers: the input layer, connected to a vector corresponding to the data set, and the output layer (or map), which is an array of node (also called neurons). The map can be also divided into so many c-planes (component planes) as data variables, representing the variable contribution to each node in the map. In the present study, the Kohonen map was a 48 unit (8×6) rectangular grid.

3. Results and discussion

The concentrations of 13 fluorinated organic compounds in whole blood from subjects living in Tarragona County (Catalonia, Spain) are summarized in Table 2. Among the 13 FOCs analyzed, seven compounds were found above the detection limit. However, only PFHxS (range: 0.65–19.96 ng/ml), PFOA (range: 0.79–3.13 ng/ml), and PFOS (range: 0.76–16.17 ng/ml) could be detected in all samples. The maximum PFHXs and PFOS levels corresponded to the same individual (ID: 10), a 24 years-old male. At the current detection limits, PFNA, PFOSA, PFDA and PFUnDA were only detected in 54.1%, 87.5%, 16.6% and 39.6% of the samples, respectively. The highest concentrations corresponded to PFOS, followed by PFHxS and PFOA.

In recent years, interest in human exposure to FOCs has led to their measurement in whole blood, plasma and serum. Recently, Ehresman et al. (in press) compared human whole blood, serum, and plasma of 18 voluntary participants for the determination of 4 FOCs. The results corroborated that whole blood concentrations of PFHS, PFOS, and PFOA are approximately half that of plasma or serum, whereas serum to plasma ratios for PFOS, PFHS and PFOA were 1:1, being the ratios independent of the level of concentrations measured. It was concluded that successful analysis of FOCs could be performed in any of the three blood matrices.

Table 3 summarizes the mean concentrations of the 13 FOCs for all 48 subjects divided according to two age groups. In general terms, for the seven detected compounds the FOC levels were higher in the group of 55±5 years. However, a significant difference was only found for PFOSA ($p<0.001$). In contrast, PFHxS concentrations were significantly higher ($p<0.05$) in the group of 25±5 years. On the other hand, the influence of gender on the concentrations of FOCs was also examined (Table 4). Significant differences were only found for PFHxS and PFOA, with males showing higher levels than females ($p<0.05$). Although the difference was not statistically significant ($p>0.05$), PFOS had a similar behavior. The current findings concerning gender are in agreement with the results reported by Kannan et al. (2004) in human blood from several countries. Similarly, no gender-related differences in the concentrations of PFOS and PFOA were reported by

Table 4
Mean concentrations (ng/ml) of 13 perfluorinated chemicals in whole blood of subjects living in Tarragona County (Catalonia, Spain) classified according to gender

	Men	Women	p
PFBuS	ND	ND	
PFHxA	ND	ND	
PFHpA	ND	ND	
PFHxS	4.58±3.80	2.55±1.23	<0.05
THPFOS	ND	ND	
PFOA	2.02±0.71	1.57±0.52	<0.05
PFNA	0.43±0.30	0.45±0.29	NS
PFOS	8.47±3.90	6.81±2.98	NS
PFOSA	0.41±0.24	0.44±0.31	NS
PFDA	0.34±0.19	0.26±0.09	NS
PFUnDA	0.33±0.19	0.36±0.19	NS
PFDS	ND	ND	
PFTDA	ND	ND	

ND: not detected. NS: not statistically significant ($p>0.05$).

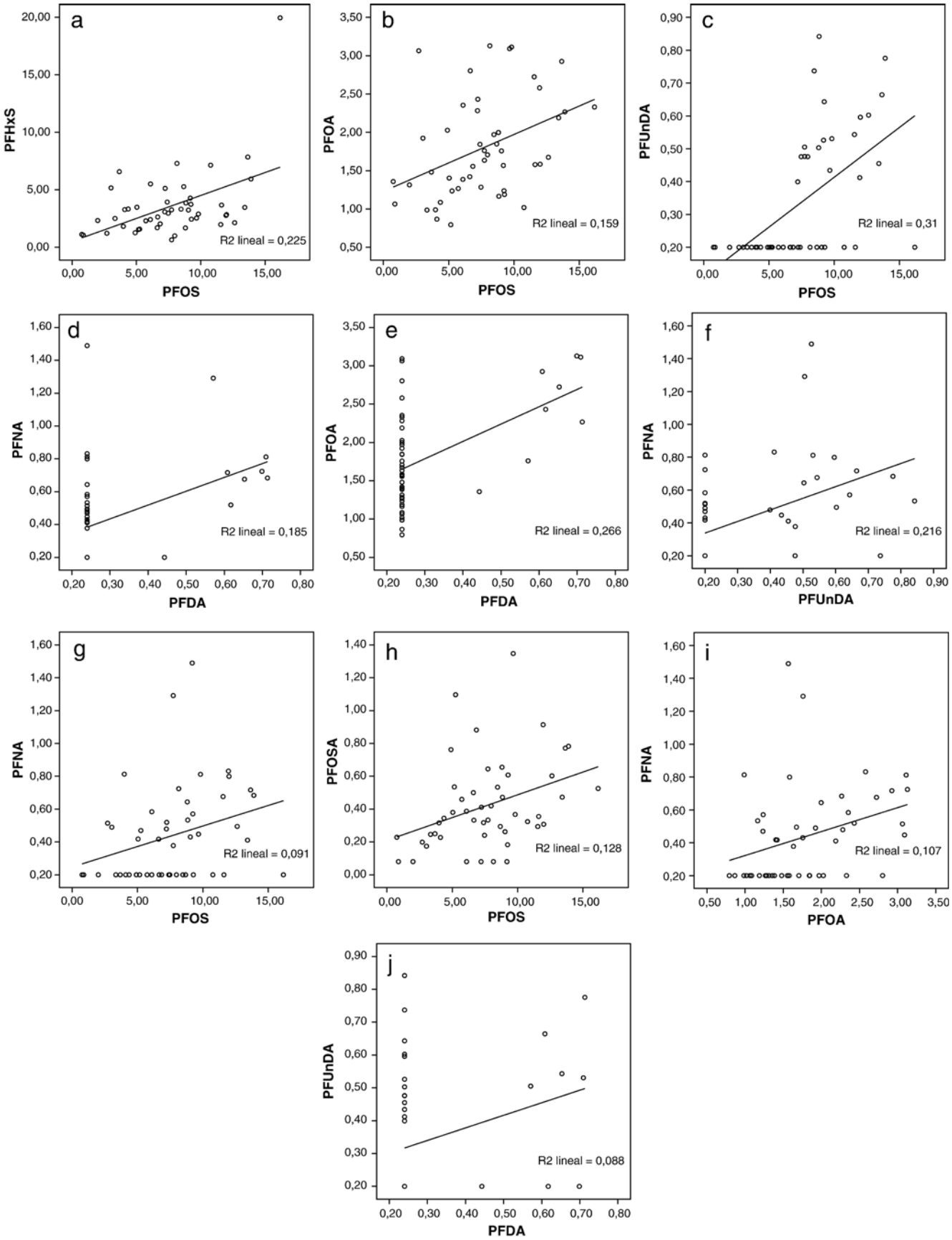

Fig. 1. Significant correlations found among a series of FOCs. From a to f: $p<0.01$; from *g* to *j*: $p<0.05$.

Olsen et al. (2003) in the serum of American Red Cross adult blood donors. The lack of clear sex-related differences noted in this study is different from the trend observed for lipophilic contaminants such as PCBs, for which concentrations in adult females are usually significantly lower than those found in adult males. Kannan et al. (2004) suggested that it could be due to the transfer of these pollutants to offspring via parturition and lactation, which would not occur with FOCs. However, in a recent study in which we determined the levels of PCDD/PCDFs in plasma samples of subjects living in the same area of the present survey (Tarragona County), the mean PCDD/PCDF concentration in females was lower than that found in males. Notwithstanding, the level of statistical significance was not reached (Agramunt et al., 2005). This difference was thought to be due to a lower PCDD/PCDF dietary intake by females of that area (Llobet et al., 2003). Kannan et al. (2004) suggested that the few (or even the lack of) age- and gender-related accumulation patterns of FOCs in humans might be related to the affinity of these compounds for lipoproteins rather than for neutral lipids (Luebker et al., 2002).

With respect to the relationships among the seven detected FOCs, Pearson's test showed a significant correlation of PFOS concentrations with those of PFHxS, PFOA and PFUnDA at $p<0.01$, and with those of PFNA and PFOSA at $p<0.05$. Only PFDA levels were not significantly correlated with those of PFOS. Other significant correlations among FOC levels corresponded to PFOA and PFDA ($p<0.01$), PFNA and PFDA ($p<0.01$), PFNA and PFUnDA ($p<0.01$), PFOA and PFNA ($p<0.05$), and PFDA and PFUnDA ($p<0.05$). The regression analysis of the associations between Pearson's correlated FOC concentrations in blood samples of 48 individuals from Tarragona County is depicted in Fig. 1.

A Self-Organizing Map (SOM) was applied to the levels of the seven detected FOCs in the 48 individual samples (Fig. 2). The results can be divided into two sections. On the one hand, the Kohonen's map indicates potential similarities of the samples according to the FOC concentrations. This map has been proved to be a clustering procedure as good as other cluster techniques (Nadal et al., 2004b). However, in the present study no special clusters could be observed in the resulting map. It means that blood samples did not group according to age or gender (Fig. 2a). On the other hand, the *c*-planes show whether or not the different FOCs have a similar behavior. In fact, there are as many *c*-planes as number of pollutants. The cell occupied by each sample is the same in the map and each one of the *c*-planes. If the *c*-planes show a similar design, it means that those pollutants will have a similar tendency in the environment. Notwithstanding, the *c*-planes did not show any association among the FOCs (Fig. 2b).

Harada et al. (2004) reported that serum levels of PFOS and PFOA of Japanese individuals were influenced by several factors, being sex and residential areas the most relevant. Males showed higher PFOS and PFOA levels than females. In a recent study on the serum concentrations of PFOS and other fluorochemicals in an elderly population from Seattle (Washington, USA), Olsen et al. (2004) did not find significant differences by sex or years of residence in Seattle, whereas PFOS concentrations were slightly lower among the most elderly. In turn, no differences between men and women were reported by Corsolini and Kannan (2004) in the concentrations of PFOS, PFHxS, PFOA and PFOSA in serum samples from Italy. Kubwabo et al. (2004) determined the levels of PFOS, PFOA and PFOSA in 56 serum samples from Canadians. No differences between males and females could be noted. The authors remarked that due to the small sample size and the sampling restricted to one geographic area, their results could not represent the distribution of measured FOCs in the overall Canadian population. Although this argument would be also valid for the results of the current study, the present data can be useful in the design of a more complete study for determining FOCs in the general population of Catalonia and Spain.

Recently, Olsen et al. (2005) investigated in blood of US citizens whether there had been a change in the concentrations of PFOS, PFOA, and five other FOCs since 1974. There were no significant differences between the 1989 and 2001 results, although the median levels of PFOS, PFOA, PFHS and PFOSA were significantly higher in 1989 (plasma samples) than in 1974 (serum samples). Data corresponding to

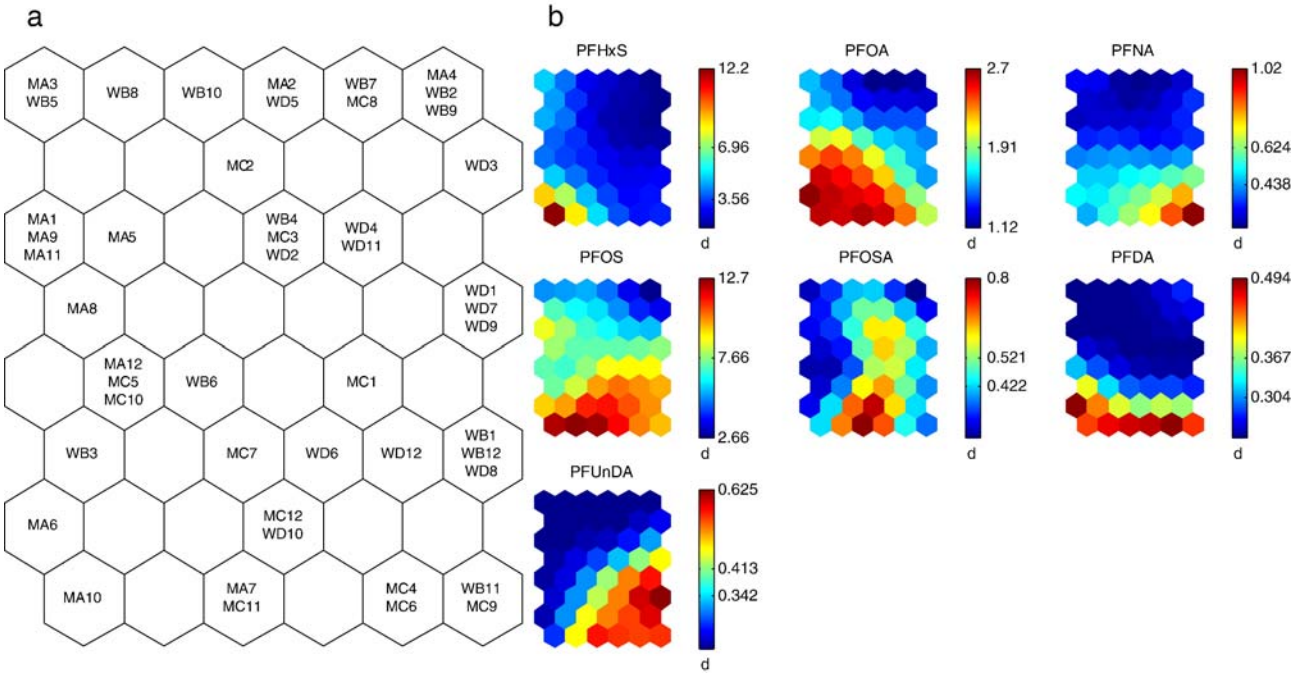

Fig. 2. Kohonen Self-Organizing Map (a) and component planes (b) for blood samples of 48 individuals. MA: young male; WB: young female; MC: adult male; WD: adult female.

1974 showed that PFOS and PFHS levels in serum were significantly higher in males than in females, which was also noted for 1989 plasma samples. Concentrations of some FOCs increased with age in both the 1974 and 1989 data. The authors indicated that comparison with other regional data collected in 2001 did not suggest a continued increase in concentrations since 1989 (Olsen et al., 2005). Guruge et al. (2005) determined concentrations and accumulation of 13 FOCs in human serum and seminal plasma in urban and rural tea worker populations from Sri Lanka. PFOS, PFHxS, PFUnA, PFDA, PFNA and PFOA were detected in all serum samples. Although these six compounds, together with PFOSA, were also detected in the current study, there were notable differences in the results. The mean PFOS and PFOA serum levels in Sri Lanka were 3.8 and 6.4 ng/ml, respectively. In Tarragona County, mean whole blood levels were 7.6 and 1.8 ng/ml, respectively, or presented on a serum basis (by multiplying whole blood concentrations by a factor of 2) (Kannan et al., 2004), 15.2 (PFOS) and 3.6 (PFOA) ng/ml.

During 2006, the results of various studies over the world have been reported. Calafat et al. (2006a) measured 11 FOCs in 23 pooled serum samples collected in the USA between 1990 and 2002, and in serum samples collected in 2003 from 44 residents in Trujillo (Peru). PFOS and PFOA were detected in all the pooled samples with median concentrations of 31.1 and 11.6 ng/ml, respectively. The frequencies of detection, and the concentration ranges of FOCs, were much lower in the samples from Peru. The same group (Calafat et al., 2006b) measured the serum concentrations of PFOS, PFOA, and 8 other FOCs in 54 pooled serum samples collected from 1832 participants of the 2001–2002 National Health and Nutrition Examination Survey (USA). Participants were 12 years of age and older. The pools represented three major racial groups/ethnicities (non-Hispanic blacks, non-Hispanic whites, and Mexican Americans), four age categories (12–19 years, 20–39 years, 40–59 years, and 60 years and older), and both genders. The concentrations of most FOCs were similar among the four age groups. The authors remarked the importance of conducting research to identify the environmental sources and pathways of human exposure to FOCs, as different patterns of human exposure were found among the population groups examined. On the other hand, Fromme et al. (2007) determined the levels of PFOS and PFOA in 356 human plasma samples collected from an adult population in Germany in 2005. Both compounds could be detected in all samples, being their concentrations significantly correlated. Median values were 12.2 and 5.3 ng/ml for PFOS and PFOA, respectively. The authors remarked that their results were in agreement with those of other recent studies in Europe, which are generally lower than those corresponding to USA and Canadian populations. Our results are also in accordance with these findings. In another recent German study, Midasch et al. (2006) determined the levels of PFOS and PFOA in plasma of 105 non-smokers adults. Median PFOS and PFOA concentrations were respectively 22.3 and 6.8 ng/ml. In 85 whole blood samples belonging to subjects living in nine cities (8 provinces) from China, Yeung et al. (2006) measured the levels of 10 FOCs. Only 3 compounds could not be detected. As in most recent studies, PFOS showed the highest concentrations (3.72–79.2 ng/ml). No age-related differences were found in PFOS, PFHxS, PFOA and PFOSA, whereas gender-related differences were noted in PFOS and PFHxS (males had higher levels than females), and in PFUnDA (females had higher concentrations than males). In turn, in a recent study Kärrman et al. (2006a) determined the levels of various FOCs in 40 pooled samples of 3802 Australian subjects. PFOS, PFHxS, PFOA, PFNA and PFOSA could be detected in all samples. Once more, PFOS was the compound showing the highest mean concentration (20.8 ng/ml), followed by PFOA (7.6 ng/ml) and PFHxS (6.2 ng/ml). An increase in PFOS concentrations with increasing age was observed in both genders. The male pool levels of some of the age groups compared to females were higher for PFOS, PFOA, and PFHxS, while in contrast, PFNA concentrations were higher in female pools. The same research group measured also the levels of 12 FOCs in whole blood samples from Sweden. The median PFOS and PFOA concentrations were 17.1 and 2.5 ng/ml (Kärrman et al., 2006b). These authors noted that Swedish residents were exposed to a large number of FOCs to the same extent than in the countries where data are currently available. However, in contrast to the results of Ehresman et al. (in press), it was indicated that the comparison of FOC levels determined in plasma with those measured in whole blood should be made with caution, as plasma/whole blood ratios between 1.1. and 1.4 were found for 4 FOCs, and a ratio of 0.2 was found for PFOSA (Kärrman et al., 2006b). Recently, Harada et al. (2007) measured the PFOS and PFOA concentrations in historically recorded human serum samples collected in Kyoto and other 10 locations throughout Japan. The historical samples collected from 1983 to 1999 demonstrated that the PFOA concentrations in males and females from Kyoto increased 4.4-fold and 4.3-fold at a rate of increase of 0.49 ng/ml/year and 0.42 ng/ml/year, respectively. In contrast, serum concentrations of PFOS reached a plateau in the late 1980s. As in our study in whole blood, the serum concentrations of PFOS and PFOA were higher in males than in females.

The present study was not intended as an epidemiological investigation, but rather its purpose was to use whole blood as a matrix to evaluate the body burden of various FOCs in a non-occupationally exposed population of Catalonia. We found that the current FOC levels are similar or even lower than the concentrations of FOCs recently reported for those European countries for which currently data are available. As with other European studies, FOC concentrations are lower than those corresponding to North America.

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